

Harry Locomotive Project 3

Single Track Safe Distance System

Two trains, one track — camera maintains safe gap automatically

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Track	Track 3 — outer single loop, both trains same direction
Train A	Front train — runs freely, no BLE needed
Train B	Rear train — BLE controlled (MAC: 60:F9:FB:49:94:C9)
Detection	Colored stickers on train roofs — ceiling camera detects both
Control	Camera measures gap → adjusts Train B speed automatically
Protocol	LionChief BLE — same as Project 1 & 2, just new MAC

QUICK START: Step 1 — Put colored stickers on train roofs. Step 2 — Run calibrate.py. Step 3 — Run main.py. Done.

HOW IT WORKS

The overhead camera watches both trains at all times. Each train has a different-colored sticker on its roof so the camera can tell them apart. Every frame, the system measures the pixel distance between the two sticker blobs. This distance maps to a speed zone, and Train B (rear) is commanded at the matching speed via Bluetooth.

Zone	Gap	Speed	What happens
DANGER	< 80px	STOP	Trains dangerously close — Train B stops immediately
WARNING	< 150px	Spd 2	Too close — minimum speed to hold position
CAUTION	< 230px	Spd 3	Closing gap — reduce speed smoothly
SAFE	< 320px	User	Comfortable following distance — user-set speed
FAR	> 450px	Spd 7	Gap too large — speed up to catch Train A

Speed Smoothing: Speed changes are smoothed — braking is faster than acceleration (safety). This prevents jerky speed changes when the gap fluctuates slightly.

Hysteresis: Zone boundaries have a small overlap so the speed does not oscillate when the gap hovers right on a threshold.

Missing train safety: If either train sticker is not visible for 2 seconds, Train B slows to a safe holding speed automatically.

SETUP — FIRST TIME ONLY

1 Activate virtual environment and install packages

In VS Code: Ctrl + ` to open terminal

→ `.venv\Scripts\activate`

→ `pip install -r single_track\requirements.txt`

2 Place colored stickers on train roofs

Train A (front, free-running): bright RED sticker on roof

Train B (rear, BLE-controlled): bright YELLOW sticker on roof

Sticker size: at least 2×2cm — must be visible from ceiling camera

Colors must be clearly different from the track and layout background

TIP: Colored sticky notes cut to size work perfectly

3 Navigate to the single_track folder

→ `cd single_track`

```
cd C:\Users\peter\Documents\Linchief_train\single_track
```

STEP 1 — CALIBRATE COLORS & DISTANCES

1 Run calibration tool

→ python calibrate.py

```
python calibrate.py
```

2 Sample Train A color

Press TAB — make sure 'Train A' is shown as active at top

LEFT CLICK directly on Train A's colored sticker in the live image

An orange circle should appear over Train A

If the blob is messy (lots of noise): press + to increase sensitivity

If the blob misses part of the sticker: press - to decrease sensitivity

3 Sample Train B color

Press TAB to switch to Train B

LEFT CLICK on Train B's sticker color in the image

A green circle should appear over Train B

Adjust with + / - until clean blob with no extra noise

4 Set distance thresholds

Position both trains at various gaps and press the matching key:

D = DANGER — bring trains very close, press D

W = WARNING — slightly further apart, press W

C = CAUTION — moderate gap, press C

S = SAFE — ideal comfortable following distance, press S

F = FAR — very wide gap where Train B should catch up, press F

The measured pixel distance is shown on screen between the trains

5 Save calibration

Press A to save all settings to calibration.json

Only needs to be done once (or if camera angle / lighting changes)

STEP 2 — RUN THE SYSTEM

1 Power on both trains

Turn on Train A (front) — it will run freely

Turn on Train B (rear) — MAC: 60:F9:FB:49:94:C9

Turn phone Bluetooth OFF — it interferes with ESP32 BLE

2 Run main.py

→ python main.py

```
python main.py
```

3 System is running

Camera window shows both trains with colored circles

A line between them shows the current gap in pixels and zone name

Top bar shows zone status and Train B speed

Train B speed adjusts automatically — no action needed

Set your preferred cruising speed with keys 1-7 in the window

KEYBOARD COMMANDS

Click on the camera window first to enable keyboard input.

1 - 7	Set Train B cruising speed (used in SAFE zone)
+ / -	Increase or decrease speed by one step
S	Manual STOP Train B
R	Manual RESUME Train B at current user speed
P	Pause / resume auto control (Train B holds last speed)
H	Toggle horn ON / OFF
B	Toggle bell ON / OFF
L	Toggle lights ON / OFF
Q / ESC	Quit — sends stop command then disconnects cleanly

TROUBLESHOOTING

Train sticker not detected	Re-run calibrate.py and click the sticker with better lighting. Increase sensitivity with +.
False detections (noise)	Increase MIN_BLOB_AREA in config.py. Reduce sensitivity in calibrate.py with -.
Train B too jerky	Reduce ALPHA_SLOW_DOWN in config.py (e.g. 0.4). Increase MIN_COMMAND_INTERVAL_MS.
Train B stops too late	Increase DISTANCE_DANGER in config.py or calibrate.py distance D threshold.
BLE not connecting	Turn phone Bluetooth OFF. Confirm TRAIN_B_MAC in config.py matches your train.
Wrong camera opens	Change CAMERA_INDEX in config.py to 0, 1, or 2.
Both trains same blob	Use more distinct sticker colors. Re-run calibrate.py.
Changing the train	Update only TRAIN_B_MAC in config.py — everything else stays the same.

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Ahmed Ali, CEO | Harry Locomotive Project 3 | June 2026 | v1.0